

DATA VISUALIZATION — SET 26 TO 30

SET 26: STUDENT MINI DATA ANALYSIS

Dataset: Student_Mini_Data.csv

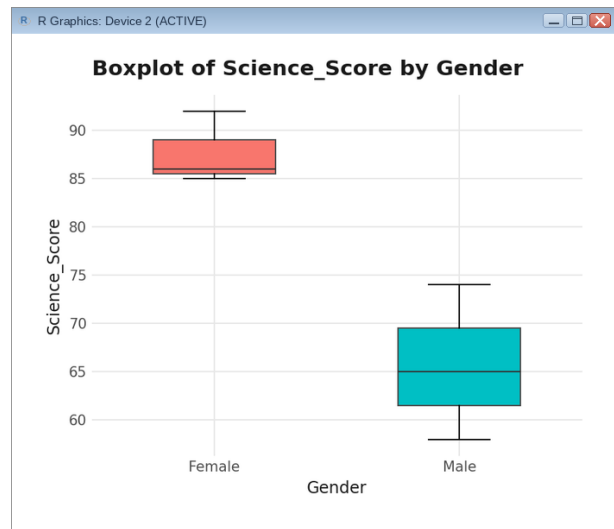
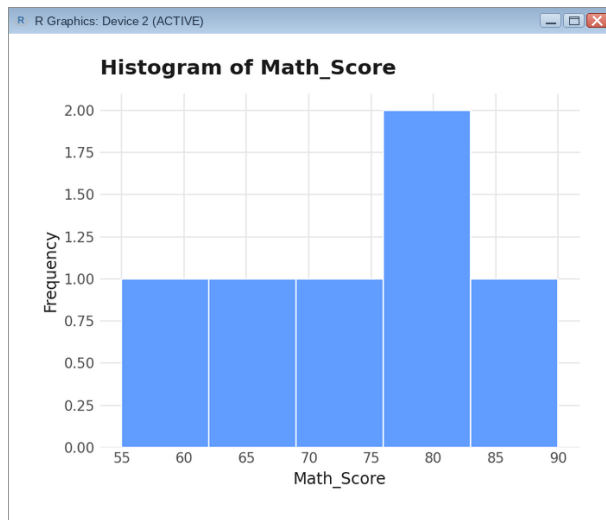
Student_ID	Gender	Study_Hours	Attendance	Math_Score	Science_Score	Exam_Date
S01	Male	2.0	78	62	65	2025-01-10
S02	Female	3.5	90	80	85	2025-01-10
S03	Male	1.5	70	55	58	2025-02-12
S04	Female	4.0	95	90	92	2025-02-12
S05	Male	2.8	85	72	74	2025-03-15
S06	Female	3.0	92	82	86	2025-03-15

Q1. Import the dataset in R. Plot a histogram for Math_Score. Draw a boxplot of Science_Score by Gender. Add proper title and labels. Interpret the results.

CODE:

```
students <- data.frame(  
  Student_ID = c("S01", "S02", "S03", "S04", "S05", "S06"),  
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),  
  Study_Hours = c(2.0, 3.5, 1.5, 4.0, 2.8, 3.0),  
  Attendance = c(78, 90, 70, 95, 85, 92),  
  Math_Score = c(62, 80, 55, 90, 72, 82),  
  Science_Score = c(65, 85, 58, 92, 74, 86),  
  Exam_Date = as.Date(c("2025-01-10", "2025-01-10", "2025-02-12",  
                        "2025-02-12", "2025-03-15", "2025-03-15"))  
)  
  
hist(students$Math_Score,  
     main = "Histogram of Math_Score",  
     xlab = "Math_Score", col = "skyblue", border = "black")  
  
boxplot(Science_Score ~ Gender, data = students,  
        main = "Boxplot of Science_Score by Gender",  
        xlab = "Gender", ylab = "Science_Score",  
        col = c("lightgreen", "lightsalmon"))
```

OUTPUT:



Q2. Create a scatter plot using R programming between Study_Hours and Math_Score. Use different colors for Gender. Add a regression line. Explain the relationship between study time and performance.

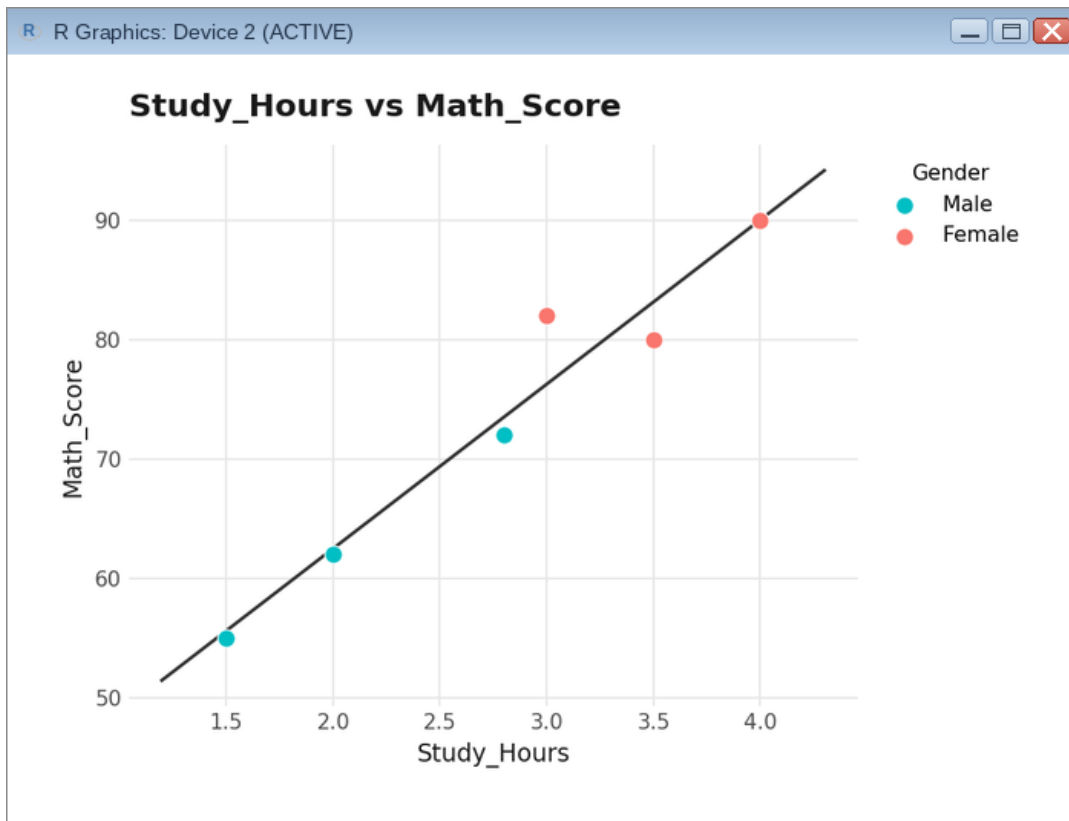
CODE:

```
plot(students$Study_Hours, students$Math_Score,
     col = ifelse(students$Gender == "Male", "blue", "red"),
     pch = 19, main = "Study_Hours vs Math_Score",
     xlab = "Study_Hours", ylab = "Math_Score")

legend("topleft", legend = c("Male","Female"),
      col = c("blue","red"), pch = 19)

model <- lm(Math_Score ~ Study_Hours, data = students)
abline(model, col = "black", lwd = 2)
```

OUTPUT:



Q3. Using R, Convert Exam_Date into Date format. Calculate monthly average Math scores. Plot a line chart for the trend. Apply moving average smoothing. Identify any increasing or decreasing pattern.

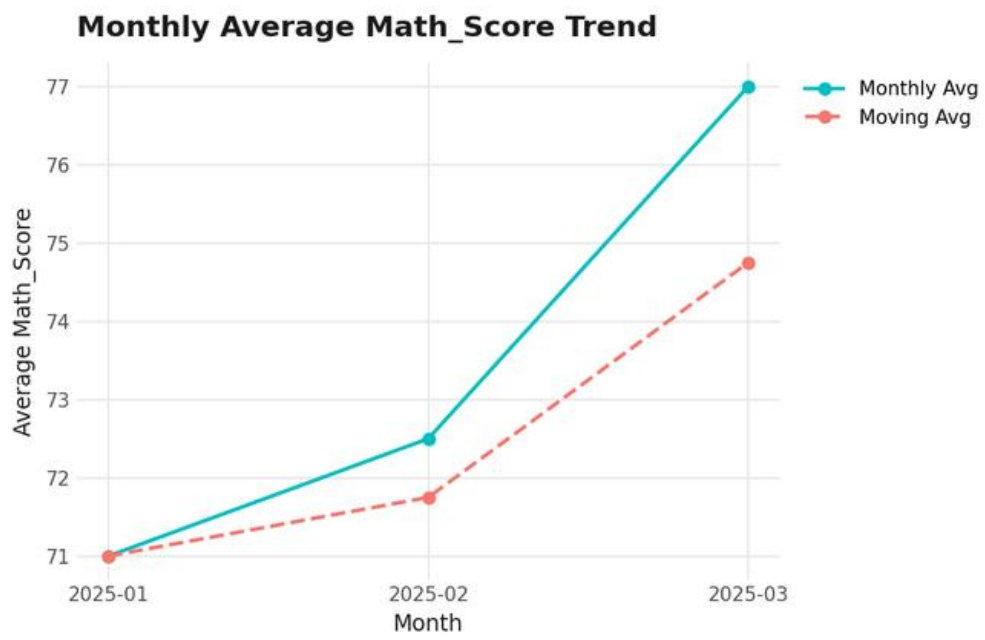
CODE:

```
students$Exam_Date <- as.Date(students$Exam_Date)
students$Month <- format(students$Exam_Date, "%Y-%m")

monthly_avg <- aggregate(Math_Score ~ Month, data = students, FUN = mean)
monthly_avg$MA <- stats::filter(monthly_avg$Math_Score, rep(1/2, 2), sides = 1)

plot(monthly_avg$Month, monthly_avg$Math_Score, type = "o", col = "blue",
     main = "Monthly Average Math_Score Trend",
     xlab = "Month", ylab = "Average Math_Score")
lines(monthly_avg$Month, monthly_avg$MA, type = "o", col = "red", lty = 2)
legend("topleft", legend = c("Monthly Avg", "Moving Avg"),
     col = c("blue", "red"), lty = c(1, 2))
```

OUTPUT:



SET 27: PATIENT HEALTH RISK ANALYSIS

Dataset: Patient Health Risk

Patient_ID	Age	BMI	BP	Cholesterol
P1	25	22	120	180
P2	40	28	135	210
P3	55	30	145	240
P4	35	26	130	200
P5	60	32	150	260

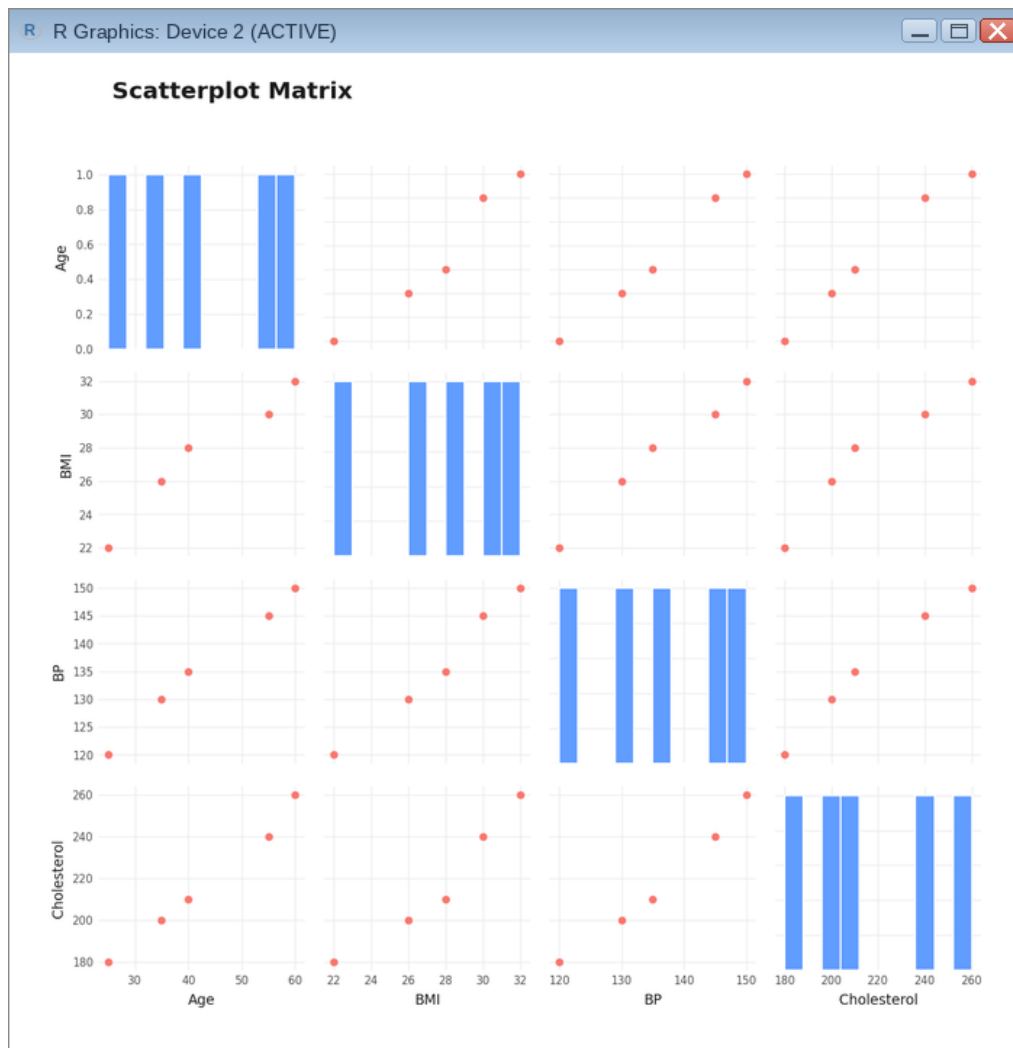
Q1. Using R, create a scatterplot matrix for Age, BMI, BP, and Cholesterol. Examine pairwise relationships to identify strong correlations, clusters, or potential risk patterns.

CODE:

```
patients <- data.frame(
  Patient_ID = c("P1","P2","P3","P4","P5"),
  Age = c(25,40,55,35,60),
  BMI = c(22,28,30,26,32),
  BP = c(120,135,145,130,150),
  Cholesterol = c(180,210,240,200,260)
)

pairs(patients[, c("Age","BMI","BP","Cholesterol")],
      main = "Scatterplot Matrix",
      pch = 19, col = "black")
```

OUTPUT:



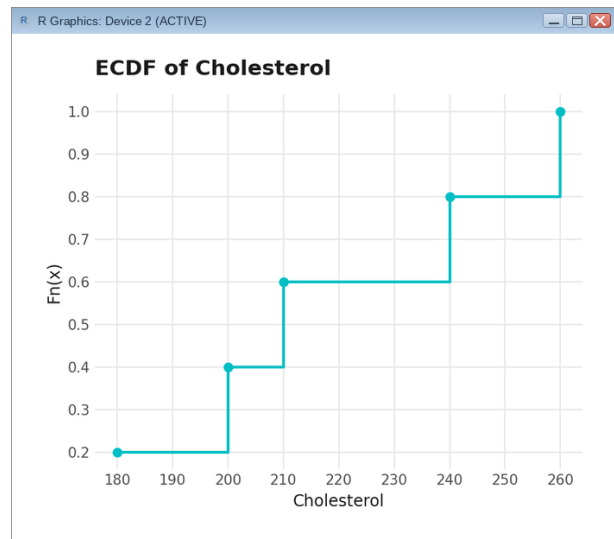
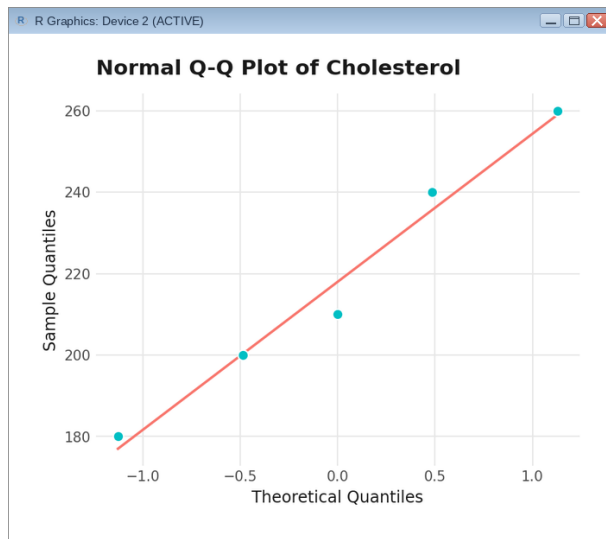
Q2. Construct Q-Q plot and ECDF for Cholesterol levels to assess distribution behaviour. Comment on normality assumption and skewness.

CODE:

```
qqnorm(patients$Cholesterol, main = "Normal Q-Q Plot of Cholesterol",
       col = "blue", pch = 19)
qqline(patients$Cholesterol, col = "red")

plot(ecdf(patients$Cholesterol),
     main = "ECDF of Cholesterol",
     xlab = "Cholesterol", ylab = "Fn(x)", col = "blue")
```

OUTPUT:



Q3. Using R, create a bar chart showing the average values of each health indicator (Age, BMI, BP, Cholesterol). Include axis labels, a descriptive title, and distinct colors for each bar.

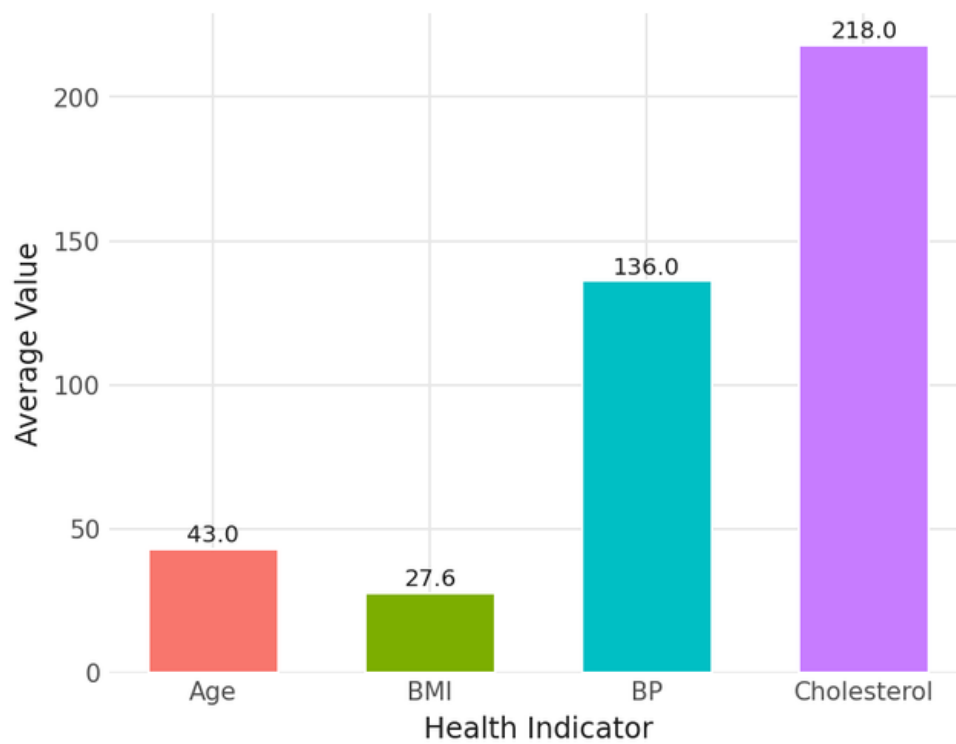
CODE:

```
avg_values <- sapply(patients[, c("Age", "BMI", "BP", "Cholesterol")], mean)

bp <- barplot(avg_values,
  main = "Average Health Indicator Values",
  xlab = "Health Indicator", ylab = "Average Value",
  col = c("steelblue", "orange", "seagreen", "tomato"))
text(bp, avg_values + 2, labels = round(avg_values, 1))
```

OUTPUT:

Average Health Indicator Values



SET 28: VEHICLE PERFORMANCE ANALYSIS

Dataset: Vehicle Performance

Vehicle_ID	Engine_Size (L)	Horsepower	Fuel_Efficiency (km/l)	Top_Speed (km/h)	Safety_Rating
V1	1.5	110	18	180	4
V2	2.0	150	15	200	5
V3	3.0	250	12	250	5
V4	2.5	200	14	220	4
V5	1.8	130	17	190	3

Q1. Using R, develop a violin plot to compare the distribution of Fuel_Efficiency across different Safety_Rating categories. Examine the spread, central tendency, and variability within each group.

CODE:

```
vehicles <- data.frame(  
  Vehicle_ID = c("V1","V2","V3","V4","V5"),  
  Engine_Size = c(1.5,2.0,3.0,2.5,1.8),  
  Horsepower = c(110,150,250,200,130),  
  Fuel_Efficiency = c(18,15,12,14,17),  
  Top_Speed = c(180,200,250,220,190),  
  Safety_Rating = c(4,5,5,4,3)  
)  
  
library(vioplot)  
vioplot(Fuel_Efficiency ~ Safety_Rating, data = vehicles,  
  col = "lightblue",  
  main = "Fuel_Efficiency by Safety_Rating",  
  xlab = "Safety_Rating", ylab = "Fuel_Efficiency (km/l)")
```

OUTPUT:

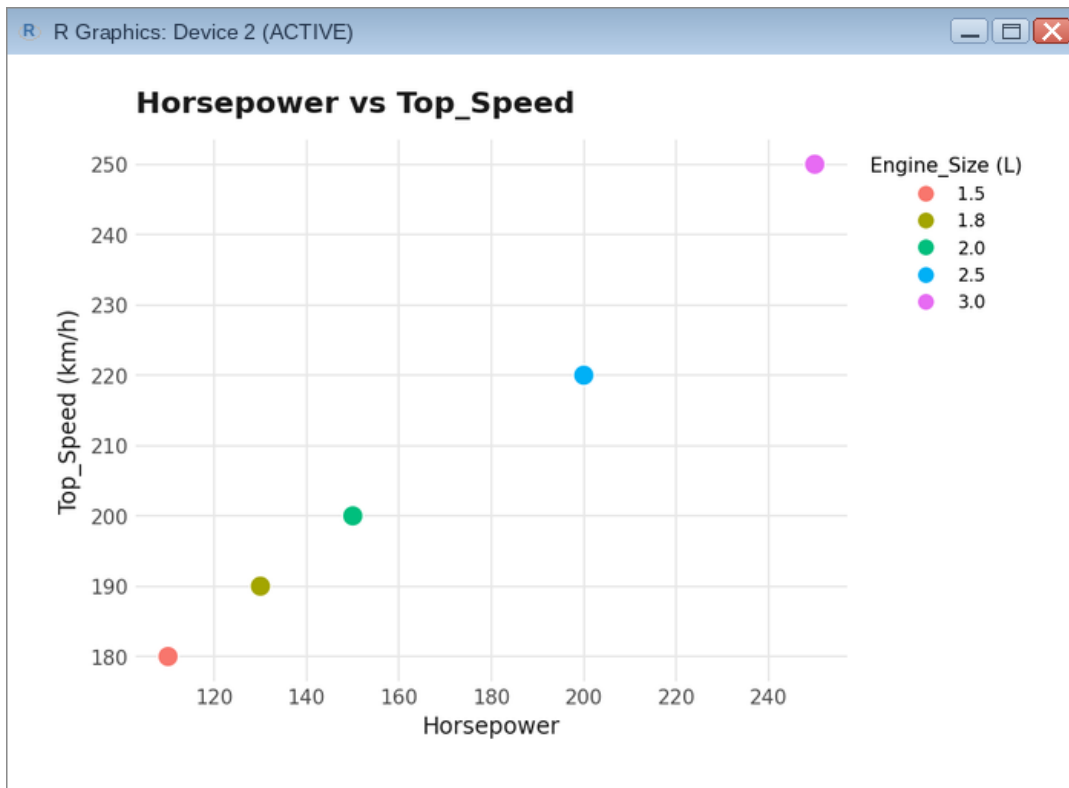


Q2. Using R, construct a scatter plot to explore the relationship between Horsepower and Top_Speed, applying color differentiation based on Engine_Size. Analyze the strength and direction of correlation.

CODE:

```
col_ramp <- colorRampPalette(c("yellow","darkgreen"))(5)
plot(vehicles$Horsepower, vehicles$Top_Speed,
     pch = 19, cex = 1.5,
     col = col_ramp[rank(vehicles$Engine_Size)],
     main = "Horsepower vs Top_Speed",
     xlab = "Horsepower", ylab = "Top_Speed (km/h)")
legend("topleft", legend = vehicles$Engine_Size,
      col = col_ramp, pch = 19, title = "Engine_Size (L)")
```

OUTPUT:



Q3. Using R, create a correlation heatmap for all numerical variables in the dataset. Identify the variables most strongly associated with Top_Speed.

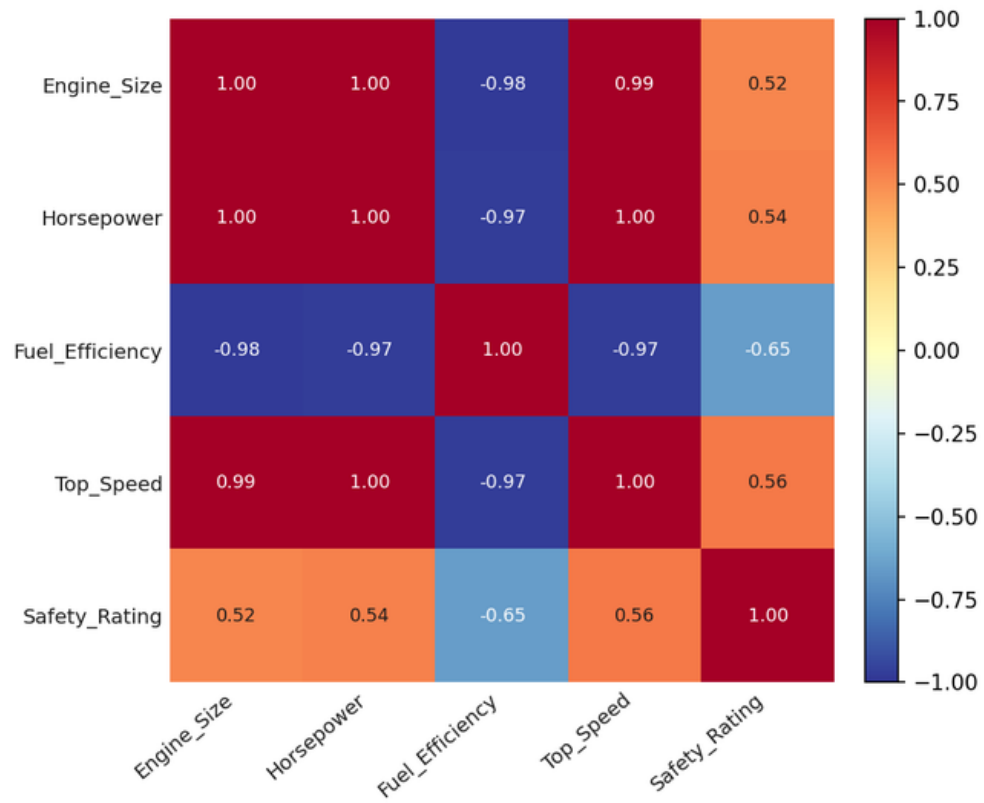
CODE:

```
library(corrplot)
num_data <- vehicles[, c("Engine_Size", "Horsepower", "Fuel_Efficiency",
                        "Top_Speed", "Safety_Rating")]
corr_matrix <- cor(num_data)

corrplot(corr_matrix, method = "color", addCoef.col = "black",
         title = "Correlation Heatmap", mar = c(0,0,1,0))
```

OUTPUT:

Correlation Heatmap



SET 29: STUDENT ACADEMIC PERFORMANCE

Dataset: Student Academic Performance

Student_ID	Age	Study_Hours	Attendance (%)	Test_Score	Participation_Score
S1	19	12	90	85	8
S2	21	8	70	70	7
S3	20	15	95	92	9
S4	22	10	85	80	8
S5	23	7	60	65	6

Q1. Using R, create a stacked area chart showing Test_Score and Participation_Score across students. Analyze trends.

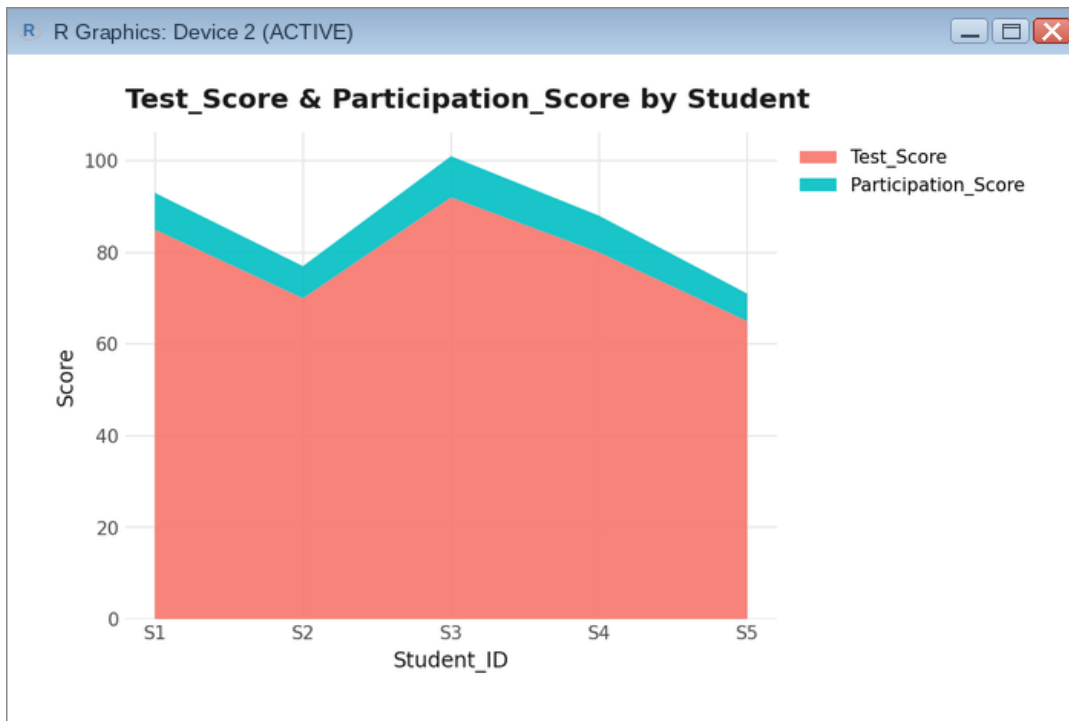
CODE:

```
students <- data.frame(
  Student_ID = c("S1","S2","S3","S4","S5"),
  Age = c(19,21,20,22,23),
  Study_Hours = c(12,8,15,10,7),
  Attendance = c(90,70,95,85,60),
  Test_Score = c(85,70,92,80,65),
  Participation_Score = c(8,7,9,8,6)
)

library(ggplot2)
library(reshape2)
df_long <- melt(students[, c("Student_ID","Test_Score","Participation_Score")],
               id.vars = "Student_ID")

ggplot(df_long, aes(x = Student_ID, y = value, fill = variable, group = variable)) +
  geom_area(position = "stack") +
  labs(title = "Test_Score & Participation_Score by Student",
       x = "Student_ID", y = "Score")
```

OUTPUT:

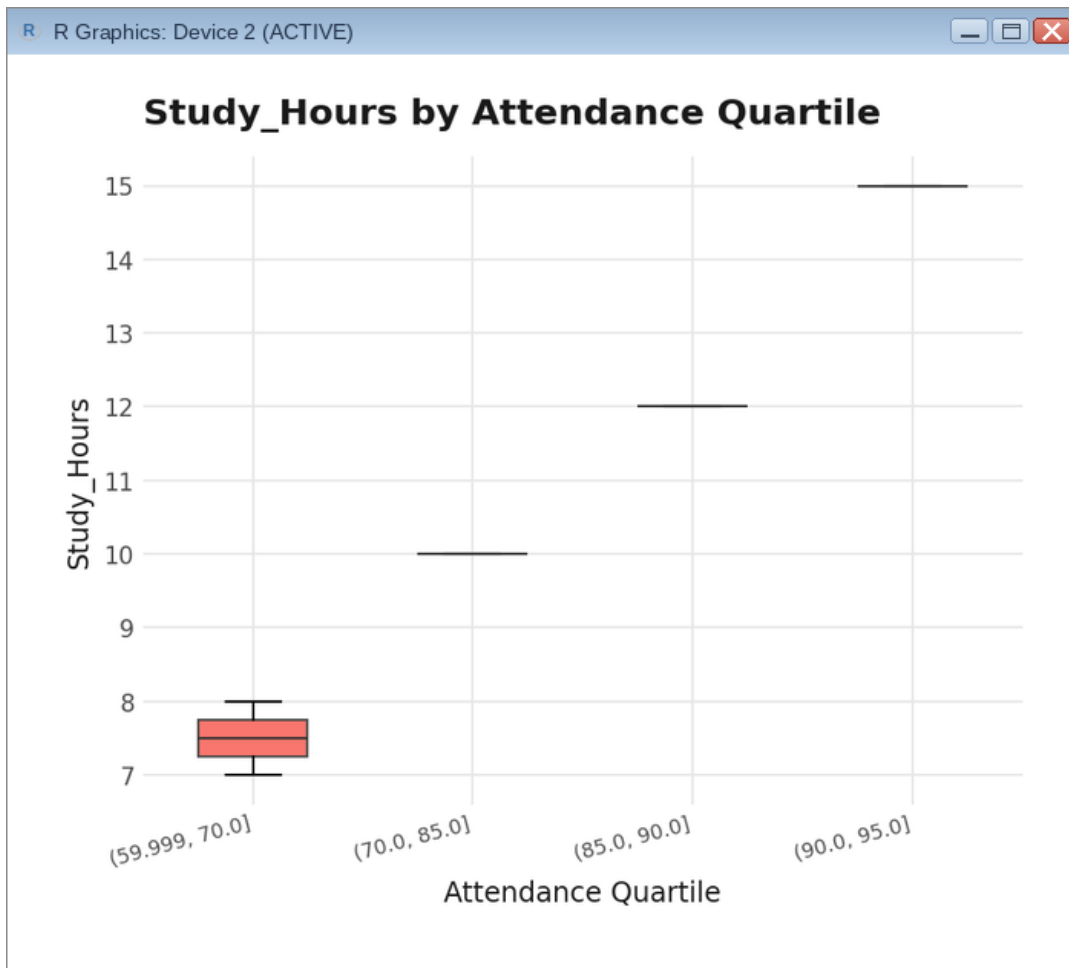


Q2. Using R, generate a boxplot of Study_Hours grouped by Attendance quartiles. Include axis labels, title, and colors for each quartile. Identify median, interquartile range, and any outliers.

CODE:

```
students$Attendance_Q <- cut(students$Attendance,  
  breaks = quantile(students$Attendance, probs = seq(0, 1, 0.25)),  
  include.lowest = TRUE)  
  
boxplot(Study_Hours ~ Attendance_Q, data = students,  
  col = c("lightyellow", "lightgreen", "lightblue", "plum"),  
  main = "Study_Hours by Attendance Quartile",  
  xlab = "Attendance Quartile", ylab = "Study_Hours", las = 2)
```

OUTPUT:



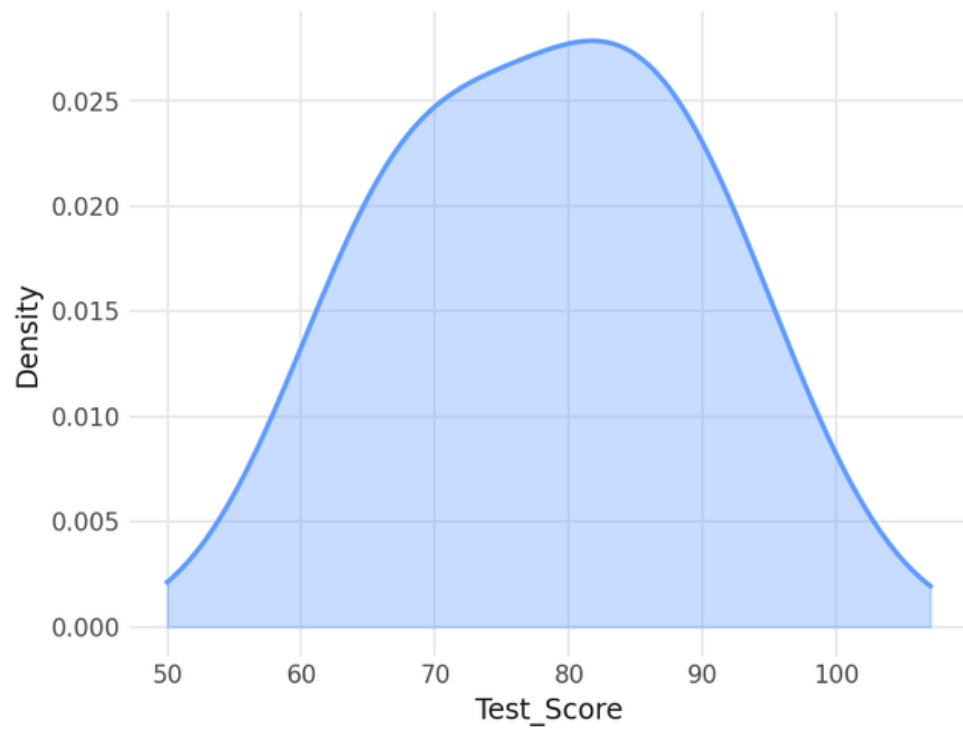
Q3. Using R, construct a density plot for Test_Score to evaluate score distribution and outliers.

CODE:

```
plot(density(students$Test_Score),  
     main = "Density Plot of Test_Score",  
     xlab = "Test_Score", col = "darkblue", lwd = 2)  
polygon(density(students$Test_Score), col = rgb(0.68, 0.85, 0.9, 0.5))
```

OUTPUT:

Density Plot of Test_Score



SET 30: MOBILE APP USAGE ANALYSIS

Dataset: Mobile_App_Usage.csv

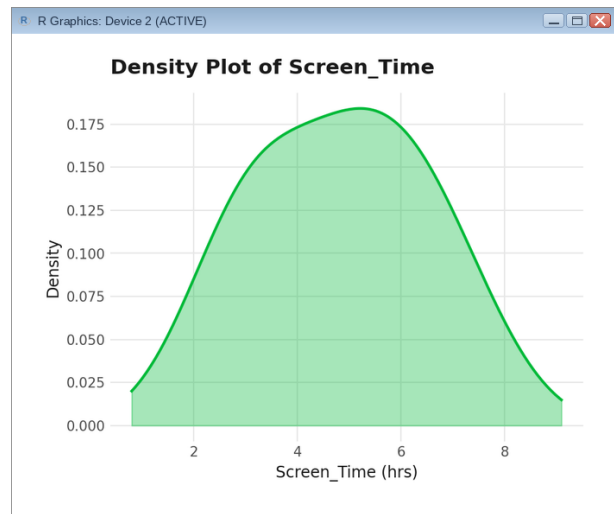
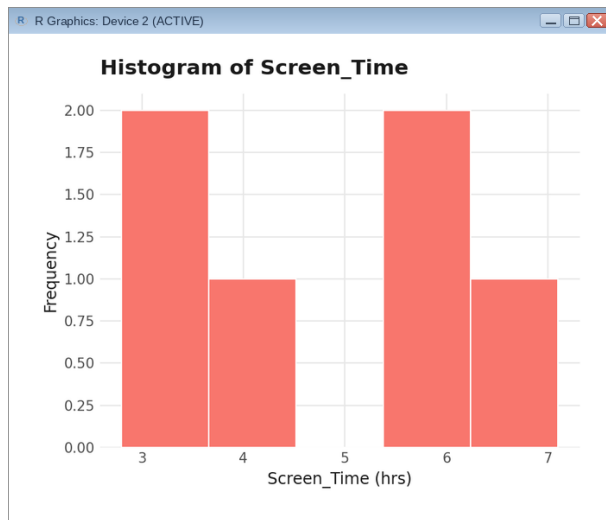
User_ID	Gender	Age	Screen_Time (hrs)	App_Usage_Count	Data_Used (GB)	Satisfaction	Usage_Date
U01	Male	20	4.5	18	2.4	3	2025-01-08
U02	Female	22	6.0	25	3.8	5	2025-01-08
U03	Male	19	3.2	12	1.6	3	2025-02-11
U04	Female	21	7.1	30	4.5	5	2025-02-11
U05	Male	23	2.8	10	1.2	2	2025-03-14
U06	Female	20	5.4	22	3.1	4	2025-03-14

Q1. Import the dataset in R. Plot a histogram and density plot for Screen_Time. Add suitable colors and labels. Interpret the screen-time distribution.

CODE:

```
usage <- data.frame(  
  User_ID = c("U01","U02","U03","U04","U05","U06"),  
  Gender = c("Male","Female","Male","Female","Male","Female"),  
  Age = c(20,22,19,21,23,20),  
  Screen_Time = c(4.5,6.0,3.2,7.1,2.8,5.4),  
  App_Usage_Count = c(18,25,12,30,10,22),  
  Data_Used = c(2.4,3.8,1.6,4.5,1.2,3.1),  
  Satisfaction = c(3,5,3,5,2,4),  
  Usage_Date = as.Date(c("2025-01-08","2025-01-08","2025-02-11",  
                          "2025-02-11","2025-03-14","2025-03-14"))  
)  
  
hist(usage$Screen_Time, col = "lightcoral", border = "black",  
     main = "Histogram of Screen_Time", xlab = "Screen_Time (hrs)")  
  
plot(density(usage$Screen_Time), col = "darkgreen", lwd = 2,  
     main = "Density Plot of Screen_Time", xlab = "Screen_Time (hrs)")  
polygon(density(usage$Screen_Time), col = rgb(0.56, 0.93, 0.56, 0.5))
```

OUTPUT:



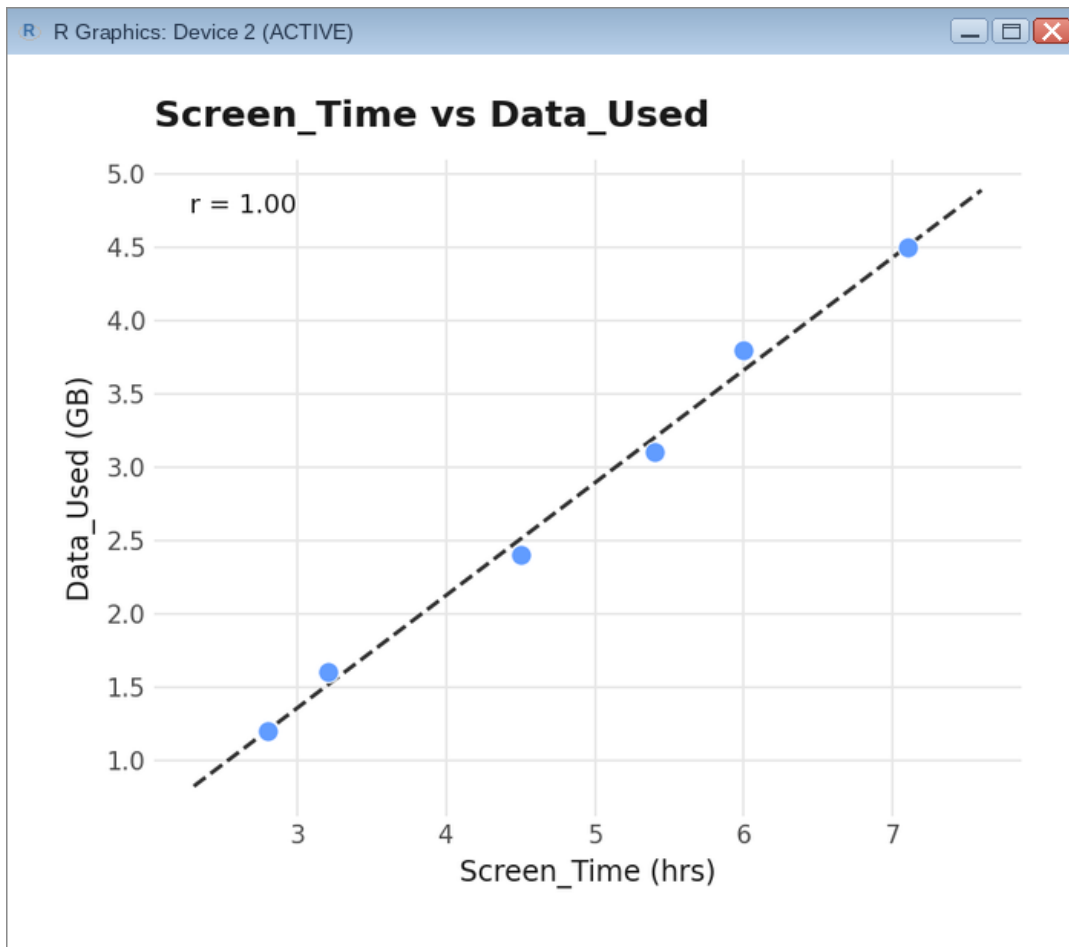
Q2. Create a scatter plot using R programming between Data_Used and Screen_Time. Calculate the correlation between them. Add a trend line. Explain the relationship.

CODE:

```
plot(usage$Screen_Time, usage$Data_Used,
     pch = 19, col = "purple",
     main = "Screen_Time vs Data_Used",
     xlab = "Screen_Time (hrs)", ylab = "Data_Used (GB)")

cor_val <- cor(usage$Screen_Time, usage$Data_Used)
model <- lm(Data_Used ~ Screen_Time, data = usage)
abline(model, col = "black", lwd = 2, lty = 2)
text(3, 4.3, paste("r =", round(cor_val, 2)))
```

OUTPUT:



Q3. Using R, find the average Satisfaction score for each Gender. Create a bar chart for the averages. Add value labels. Interpret user satisfaction patterns.

CODE:

```
avg_sat <- aggregate(Satisfaction ~ Gender, data = usage, FUN = mean)

bp <- barplot(avg_sat$Satisfaction, names.arg = avg_sat$Gender,
  col = c("lightpink","lightblue"),
  main = "Average Satisfaction by Gender",
  xlab = "Gender", ylab = "Average Satisfaction")
text(bp, avg_sat$Satisfaction + 0.1, round(avg_sat$Satisfaction, 2))
```

OUTPUT:

Average Satisfaction by Gender

